SCAPE · SEEING THE WORK · NSTA BOSTON 2027

SCAPE Free Tools: Checklist Template and Quality Criteria

Everything in this handout is free to copy and use in your classroom. It collects the tools from the Seeing the Work poster and session packet in one file.

WHAT IS INSIDE

1. Self Check Checklist Template. A blank checklist for any multi-step activity, with one row per step and a space for the icon that names the work.

2. Checklist Quality Criteria. Six tests for steps students can actually follow. If a step fails one, rewrite the step, not the student.

3. The Eight Academic Icons. The icon overview page, so the icons on your slides, sheets, and visual agendas match ours.

Printed copies of the template and criteria are on the poster table at the session. This file is the same tools in digital form, ready to print for your own classroom.

Built by the Quabbin Regional School District SCAPE team through the Massachusetts DESE SCAPE Institute, offered with Academic Discoveries. The academic icons began as student work in Quabbin's graphic design class and were refined by the SCAPE team.

TRY IT NEXT WEEK PACKET

Self Check Checklist Template

Copy freely. One checklist per activity. Steps keyed to icons.

Activity: _____

Class: _____

Date: _____

icon	_____	☐
icon	_____	☐
icon	_____	☐
icon	_____	☐
icon	_____	☐
icon	_____	☐
icon	_____	☐
icon	_____	☐
icon	_____	☐
icon	_____	☐

Finished every step? Show a partner, then initial here: _____ A checklist works when a student can finish without asking the teacher what comes next.

TRY IT NEXT WEEK PACKET

Checklist Quality Criteria

Six tests. If a step fails one, rewrite the step, not the student.



STEP 5
TRADE AND DEFEND

1 **Start with a verb.**
Every step begins with something a student can watch themselves do: read, build, count, write, trade.

2 **One action per step.**
If a step contains the word and, it is probably two steps wearing a trench coat. Split it.

3 **End in a state you can see.**
A finished step leaves evidence: a written sentence, a built strand, a number in a table.

4 **Use words students know.**
Read the step aloud. If it sounds like the teacher guide, rewrite it in student language.

5 **Key each step to its icon.**
The icon tells students what kind of work the step is before they read a word.

6 **Short enough to finish.**
Seven to ten steps is the sweet spot. Twenty steps is a second assignment, not a checklist.

WATCH IT FAIL, THEN PASS

Weak: Understand the model and be sure it is right.

Strong: Build the strand. Check it against the answer flap. Fix any mispairings.



KEYED: MODEL

TRY IT NEXT WEEK PACKET

The Eight Academic Icons

One icon for each Science and Engineering Practice.



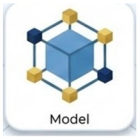
STEP 3
MEET THE ICONS



Question

SEP: Asking Questions

Wonder, notice, and ask questions that science can answer.



Model

SEP: Developing and Using Models

Build or draw a representation that shows how something works.



Investigate

SEP: Planning and Carrying Out Investigations

Gather data through a planned test or careful observation.



Analyze

SEP: Analyzing and Interpreting Data

Look for patterns in the data you collected.



Calculate

SEP: Mathematics and Computational Thinking

Measure, count, graph, and compute.



Explain

SEP: Constructing Explanations

Say how or why something happens, using evidence.



Justify

SEP: Engaging in Argument from Evidence

Defend a claim with evidence and reasoning.



Communicate

SEP: Obtaining, Evaluating, and Communicating Information

Share findings so others can use them and check them.

The icons live on slides, worksheets, station cards, and the visual agenda, so a student can tell at a glance what kind of scientific work a task is asking for.